AKHIL KUMAR PERAM

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Education

Program	Institution	%/CGPA	Year
M.Tech in Computational	Indian Institute of Technology Madras	8.47	2023 - 2025
M.Tech in Aerospace Engineering	Indian Institute of Technology Kanpur	7.25	2021 - 2023
B.Tech. in Mechanical Engineering	Hindustan Institute of Technology	7.84	2016 - 2020
XIIth std. - BIEAP	Sri Chaitanya JR College	94.6%	2014 - 2016
Xth Std. - BSEAP	Ratnam EM High School	8.8	2014

Technical Skills

Languages/ Tools : Python, SQL, MATLAB, Excel, GitHub

Tools/Frameworks: Numpy, Pandas, Matplotlib, Seaborn, Scikit-Learn, Tensorflow, NLTK, Langchain, LangSmith, Streamlit

Core Topics: Machine Learning, Deep Learning, NLP, LLM, Large Vision Model (LVM), CLIP, Gen AI, Crew AI, Autogen, MCP, ACP, ChatGPT Enterprise, OpenAI-SDk, Gemini-ADk

Experience

Technical Lead - GenAI || *HCLTech* **Aug 2025 – Present**

- Designed and deployed multiple GenAI use cases and automation solutions using **ChatGPT Enterprise, OpenAI SDK, and Gemini ADK**, accelerating workflows.
- Developed LLM-powered autonomous agents with tool calling, memory, API integrations, and retrieval workflows, enabling systems to execute tasks with minimal human intervention.

Knowledge Assistant

- Developed a full RAG-based chatbot system using FastAPI, integrating **Azure OpenAI** for LLM inference and **ChromaDB** for vector storage, enabling real-time document-grounded Q&A.
- Built a dynamic document ingestion and embedding pipeline, supporting PDF/TXT/DOCX formats and automated retriever reloading without server restart to ensure instant knowledge updates.
- Implemented context-aware conversational memory and prompt-engineering rules, improving answer precision and reducing hallucination by enforcing strict context-bound responses.

RAG-based AI Assistant with Multi-Document Switching

- Built an end-to-end RAG application using **LangChain, Azure OpenAI and ChromaDB**, including custom text chunking, metadata enrichment, and per-document collections for accurate retrieval.
- Implemented intelligent follow-up question generation and a configurable system prompt, improving answer relevance and conversational flow while strictly grounding responses in retrieved document context.
- Designed dynamic document switching in the **RAG pipeline**, allowing users to select or change the underlying knowledge base at query time and ask context-aware questions across different document sets.
- Developed a **FastAPI** backend with REST endpoints for multi-file upload, automatic ingestion into a vector store, and a interface for question-answering over selected collections.

Multimodal Document Intelligence System

- Designed and implemented a multi-modal RAG pipeline that extracts and embeds text, tables, and images separately from PDF documents using **NVIDIA** embeddings and Chroma vector stores for high-precision retrieval.
- Developed an image understanding layer by integrating **Gemini Vision** to classify document images (graphs, tables, diagrams, pure images) and convert them into searchable semantic descriptions.
- Engineered a scalable FastAPI backend with endpoints for document upload, vector embedding, and conversational querying using ensemble retrievers to boost response precision.

Insurance Claim Automator

- Designed an insurance claim automation workflow by orchestrating multi-agent collaboration using **CrewAI, ACP-SDK, and Gemini LLM**, enabling end-to-end claim document processing and email notifications.
- Integrated custom toolchain (FileProcessor, Claim History Tools & Email Sender) into agent pipelines to support document classification, claim data extraction, validation, history update, and automated reporting.

- Implemented asynchronous ACP server client communication using Server.run, Client.run sync, and asyncio to trigger execution of a sequential CrewAI process remotely with real-time response handling.

Code Agent

- Built an autonomous vibe-coding agent system where LLMs collaboratively plan, generate, and write complete applications end-to-end using tool-calling, manifest validation, and structured file-creation workflows.
- Integrated Gemini with a FastAPI control layer to trigger agent conversations and automatically package generated codebases into downloadable artifacts.
- Developed a hands-off code generation pipeline that transforms a natural-language prompt into a fully runnable project with dependency files, README, env scaffolding, and production-grade structure — without human intervention.

Runbook Incident Resolution Assistant

- Developed a fine-tuned **Gemma SLM** using Unsloth + QLoRA to transform raw runbook documents into a context-aware assistant that provides actionable troubleshooting steps, rollback plans, and department-wise POC contacts.

Data Science Intern || *My Analytics School* **June 2024 – July 2024**

- Built a RAG chatbot with **LangChain, Ollama framework, ChromaDB** as Vector database and LLMs for embeddings.
- Enhanced chatbot performance with cross-encoder as a Re-ranker and achieved an accuracy of **90%**.
- Reduced the need for a large customer service team, resulting in significant operational cost savings.

AI-Powered Teaching Assistant for Engineering Concepts || *MTech Project* **Aug 2024 – Apr 2025**

- Designing a chatbot that integrates **LLM** models to process text and multimedia inputs for interactive learning.
- Built a multimodal bot using **Python, TensorFlow, Hugging Face, and CLIP** for visual data interpretation, with a Flask backend and a responsive **HTML, CSS-based UI**.
- Implementing NLP techniques to tune the bot for clarify doubts, and apply concepts to real-world design challenges.

Projects

Evaluating Fracture Parameters from Photoelastic Images with CNN | *ML in MS* (*Course Project*) **Jan'24 – Feb'24**

- Engineered a robust Custom CNN architecture in keras that processed 10,000 training images to determine SIF values.
- Utilized Isochromatic Photoelastic images to calculate the fracture parameters for zero degrees crackangle.
- Achieved accuracy of 82% on the test data, demonstrating the effectiveness of the CNN model inidentifying the fracture.

Fradulent Credit Card Transaction Detection | *ML Personal Project* **Jan'24 – Feb'24**

- Visualized and Pre-processed a highly imbalanced dataset, conducted an EDA with **Univariate and Bivariate Analysis**.
- Applied Machine Learning algorithms such as **Logistic Regression, Support Vector Machine and K-Nearest Neighbors** and performed GridSearchCV to optimize model parameters, achieving a model classification recall up to **93%**.
- Evaluated performance with **F1 score, Confusion Matrix, and ROC curves**, found Logistic Regression as best model.

Sentiment Analysis using NLP and Machine Learning Techniques | *NLP Project* **June'24-Aug'24**

- Integrated of 5k+ movies with ratings of 100k+ from multiple csv files into a Single DataFrame.
- Implemented text cleaning with **Vectorization, Stemming, Stopwords and Lemmatization** using NLTK for Analysis.
- Employed **CountVectorizer and TF-IDF** vectorizer for text data, enhancing ML Logistic regression model accuracy.
- Predicted user rating with RMSE score of **0.89** for a movie by incorporating SVD.

CERTIFICATIONS

Machine Learning : Linear Regression, Logistic Regression, Decision Tree, SVM, KNN, Ensemble Techniques, Clustering etc.

Deep Learning : Perceptron, ANN, CNN, RNN, LSTM, NLP, Transformer.

Position of Responsibilities

AM Placement Core: Lead a 3-tier team of 7 members and 100+ students ensure effective management in placements.

AM Deputy Placement Coordinator: Coordinated placements of academic year 2023-24 for Airbus, Northern Trust and Meril Healthcare with help of AM team.

MTech Legislator: Serving as the elected representative for the M.Tech student body within the prestigious SLC at IIT Madras, democratically elected by an overwhelming mandate from over 1300+ M.Tech student

Teaching Assistant: Mentored 230+ students in the Strength of Materials Lab at IIT Madras as a Teaching Assistant.

Extra Curriculars

- Completed German Language **A1** certificate from **Goethe Zentrum**
- Represented AM department and Won gold in Volleyball PG tournament at IIT Madras